Lecture – 10

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In this lecture…

i. A Model of the Image Degradation/Restoration Process
ii. Types of Noise (White Noise, Impulse Noise - Salt & Pepper Noise, Gaussian Noise, Rayleigh Noise)

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A Model of the Image Degradation/Restoration Process

- Image degradation is modeled as an operator $\mathcal{H}$ that, along with additive noise η(x, y), affects an input image f(x, y) to produce a degraded image g(x, y).
- Restoration aims to estimate the original image $\hat{f}$(x,y) as accurately as possible.
- When $\mathcal{H}$ is linear and position-invariant, the degradation process can be represented using convolution in the spatial domain and multiplication in the frequency domain.
- Degradation is expressed as:

$$g(x,y) = (h \star f)(x,y) + \eta(x,y) \qquad \text{in spatial domain}$$

$$G(u,v) = H(u,v).F(u,v) + N(u,v) \qquad \text{in frequency domain}$$

where h(x, y) is the degradation function and η(x, y) is additive noise.

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A Model of the Image Degradation/Restoration Process

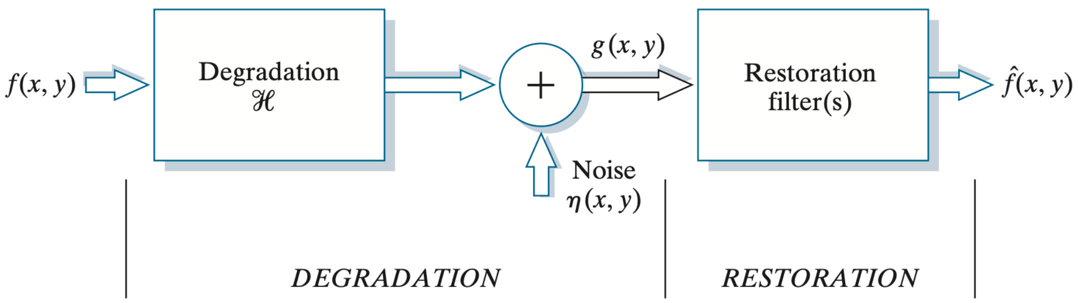

Figure: A model of the Image degradation/ Restoration process.

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Noise Models

- Noise in digital images mainly originates from environmental and sensor-related factors during acquisition, and from interference during transmission.
- Noise model assumptions:
  - Independent of Spatial co-ordinates
  - Uncorrelated with the image – no correlation between pixel values and the noise components

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White Noise

- Fourier spectrum of noise is constant. i.e. it distributes energy uniformly across the frequency domain.

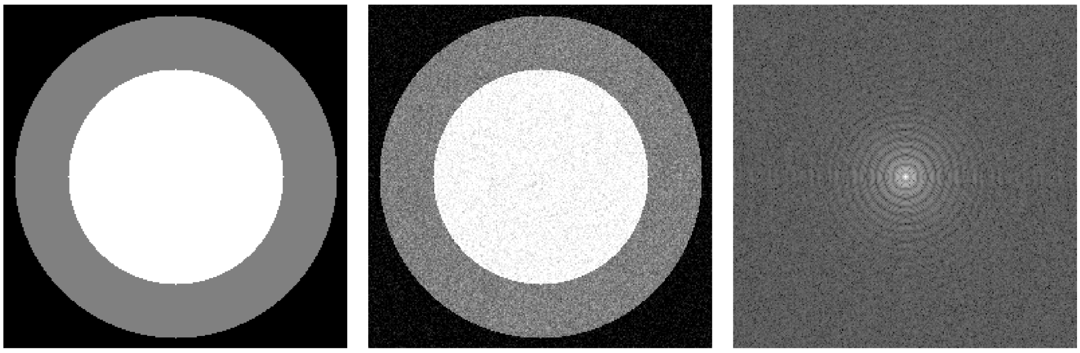

Figure: Sample Image, White Noise added image and Power Spectrum of Noisy image

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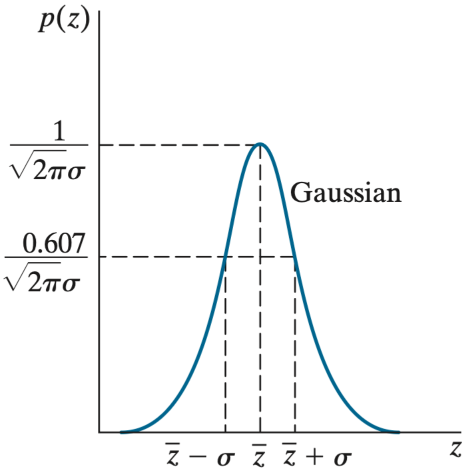
Gaussian Noise

- The PDF of a Gaussian random variable, z, is defined by the following expression:

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\bar{z})^2}{2\sigma^2}} \qquad -\infty < z < \infty$$

where z represents intensity, $\bar{z}$ is the mean (average) value of z, and σ is its standard deviation.



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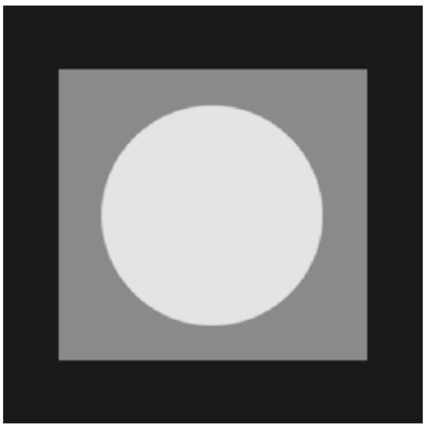
Gaussian Noise



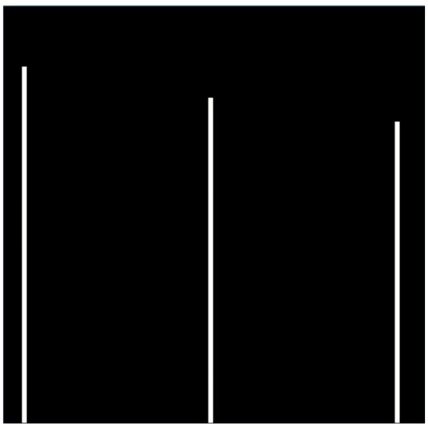


Figure: Test Image and its Histogram Plotted

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Gaussian Noise

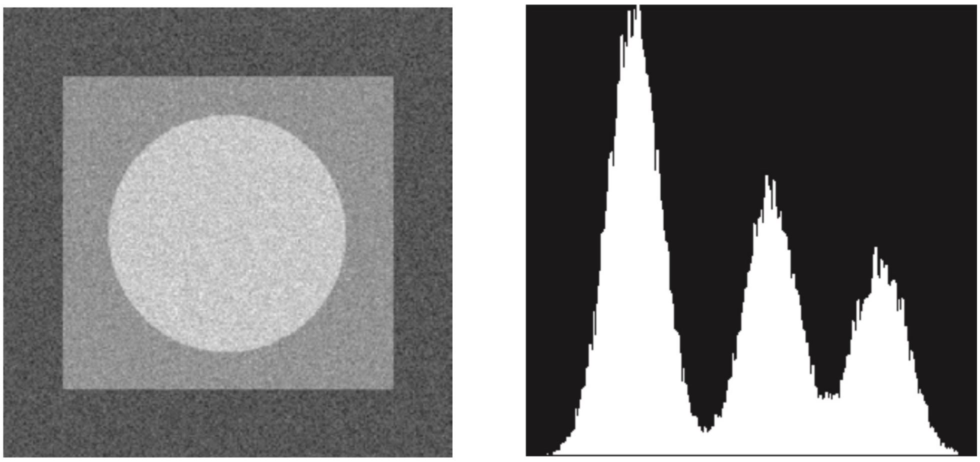

Figure: Images and histograms resulting from adding Gaussian Noise

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Rayleigh Noise

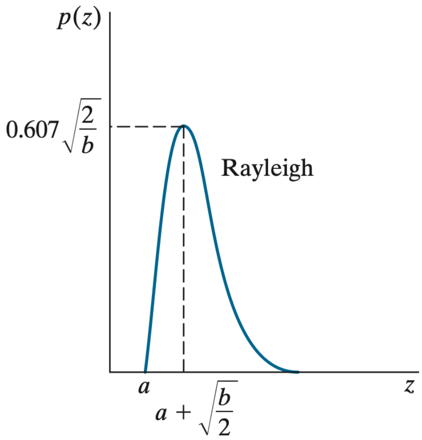
- The PDF of Rayleigh noise is given by

$$p(z) = \begin{cases} \frac{2}{b}(z - a)e^{-(z-a)^2/b} & z \geq a \\ 0 & z < a \end{cases}$$

- The mean and variance of z when this random variable is characterized by a Rayleigh PDF are

$$\bar{z} = a + \sqrt{\pi b/4}$$

$$\sigma^2 = \frac{b(4 - \pi)}{4}$$



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Rayleigh Noise

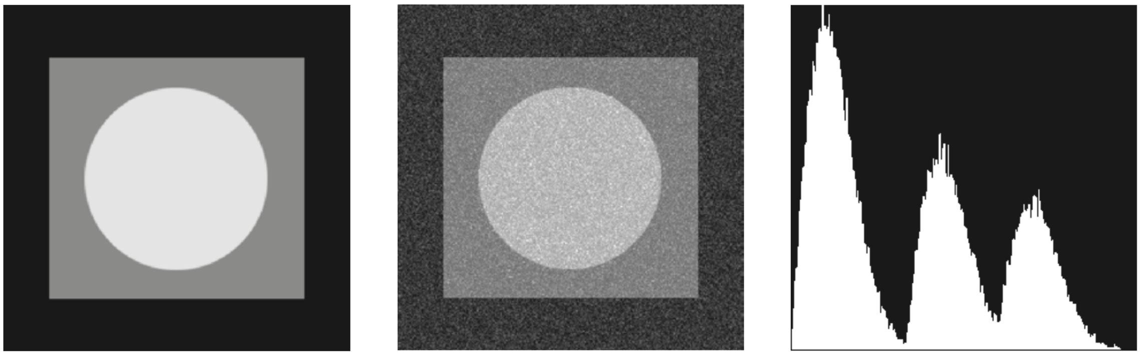

Figure: Original Image, Noise added image and histograms resulting from adding Rayleigh Noise

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Erlang (Gamma) Noise

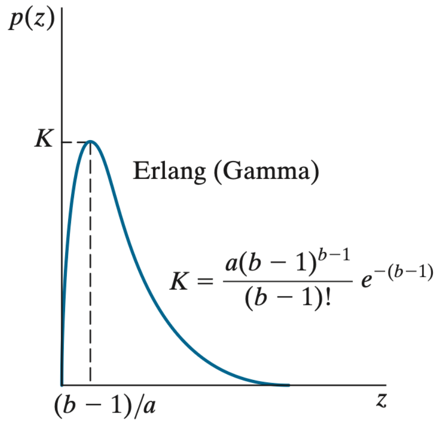
- The PDF of Erlang noise is

$$p(z) = \begin{cases} \dfrac{a^b z^{b-1}}{(b-1)!} e^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}$$

where a>b, b is a positive integer, and "!" indicates factorial.

- The mean and variance of z are

$$\bar{z} = \frac{b}{a} \qquad \sigma^2 = \frac{b}{a^2}$$



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Erlang (Gamma) Noise

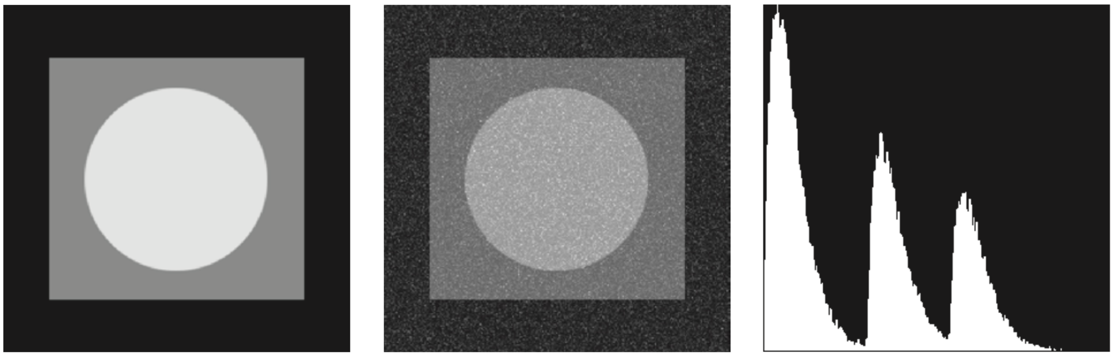

Figure: Original Image, Noise added image and histograms resulting from adding Erlang (Gamma) Noise

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Exponential Noise

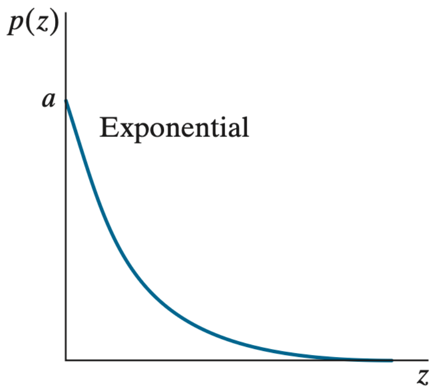
- The PDF of exponential noise is given by

$$p(z) = \begin{cases} ae^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}$$

where a > 0.

- The mean and variance of z are

$$\bar{z} = \frac{1}{a} \qquad \sigma^2 = \frac{1}{a^2}$$



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Exponential Noise

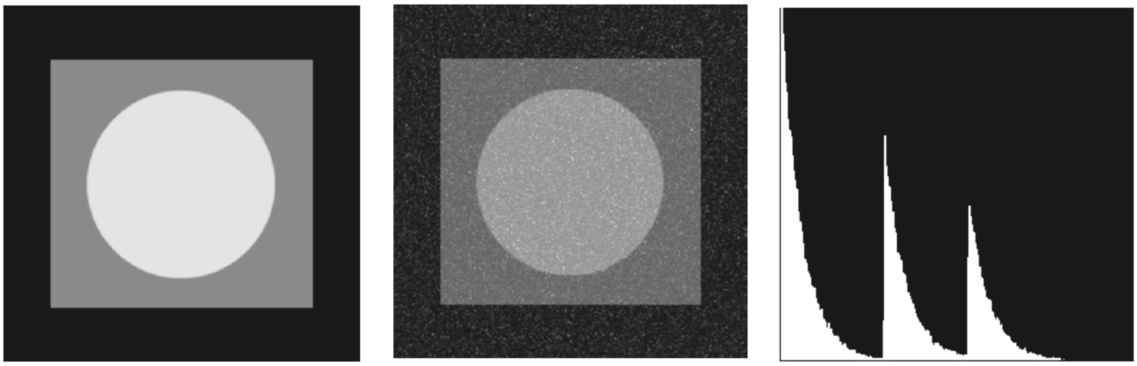

Figure: Original Image, Noise added image and histograms resulting from adding Exponential Noise

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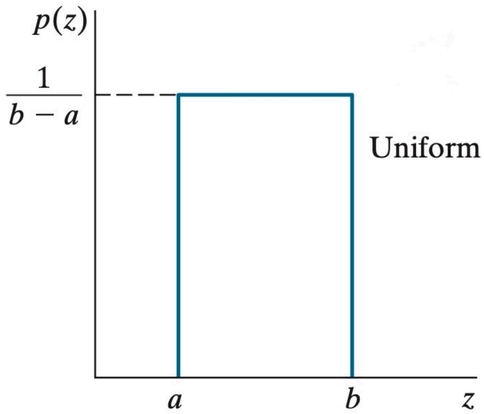
Uniform Noise

- The PDF of uniform noise is

$$p(z) = \begin{cases} \dfrac{1}{b-a} & a \le z \le b \\ 0 & \text{otherwise} \end{cases}$$

- The mean and variance of z are

$$\bar{z} = \frac{a+b}{2} \qquad \sigma^2 = \frac{(b-a)^2}{12}$$



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Uniform Noise

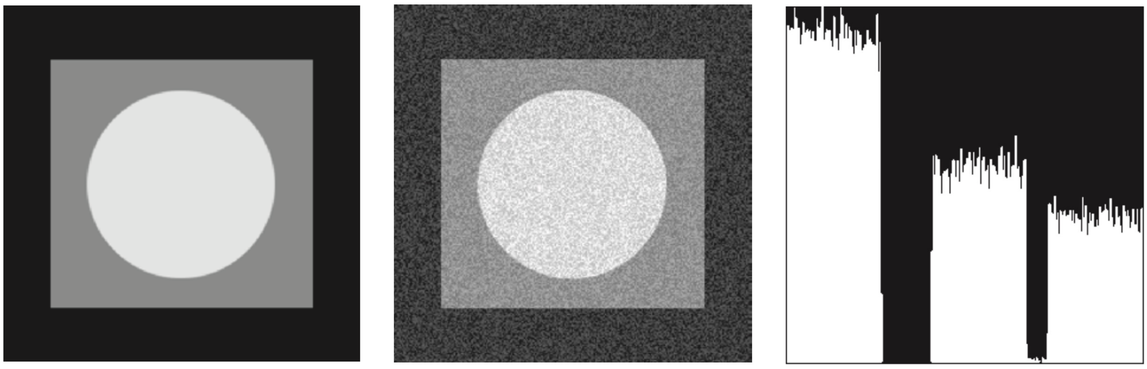

Figure: Original Image, Noise added image and histograms resulting from adding Uniform Noise

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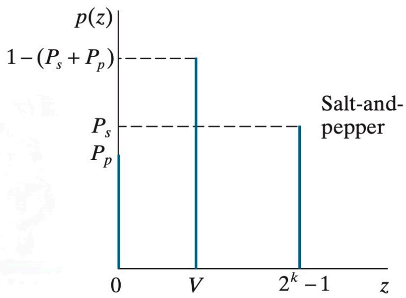
Salt & Pepper Noise

- The PDF of salt-and-pepper noise is given by

$$p(z) = \begin{cases} P_s & \text{for } z = 2^k - 1 \\ P_p & \text{for } z = 0 \\ 1-(P_s + P_p) & \text{for } z = V \end{cases}$$

k is no of bits used to represent the intensity values



- The mean of salt-and-pepper noise is given

$$\bar{z} = (0)P_p + K(1 - P_s - P_p) + (2^k - 1)P_s$$

- The variance is given by:

$$\sigma^2 = (0-\bar{z})^2 P_p + (K-\bar{z})^2(1 - P_s - P_p) + (2^k - 1)^2 P_s$$

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Salt and Pepper Noise

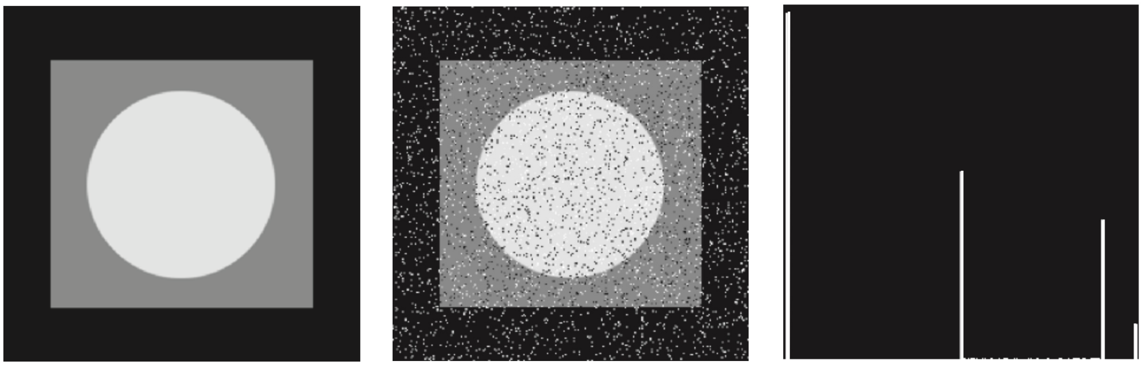

Figure: Original Image, Noise added image and histograms resulting from adding Salt and Pepper Noise

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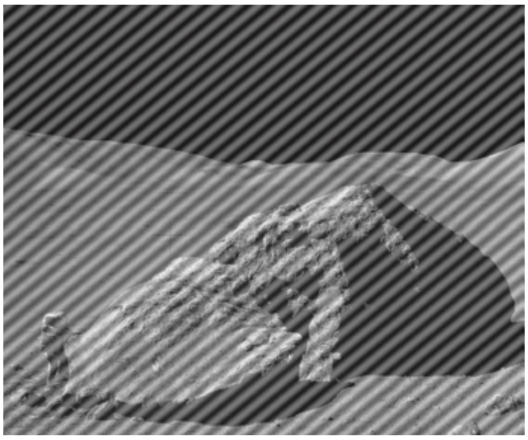
Periodic Noise

- Periodic noise in images typically arises from electrical or electromechanical interference during image acquisition.
- This can be reduced significantly via frequency domain filtering.
- Given image is corrupted by additive (spatial) sinusoidal noise.



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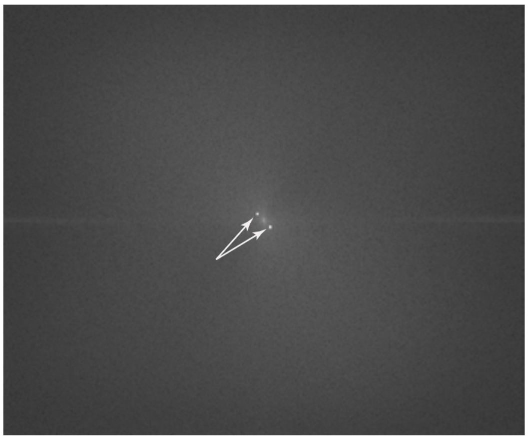
Periodic Noise

- The Fourier transform of a pure sinusoid is a pair of conjugate impulses located at the conjugate frequencies of the sine wave.
- if the amplitude of a sine wave in the spatial domain is strong enough, we would expect to see in the spectrum of the image a pair of impulses for each sine wave in the image.



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Discussions and Classwork

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Lecture – 11

Harimohan Khatri

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In this lecture…

i. Noise Models Restoration in the Presence of Noise-Only Spatial Filtering
ii. Periodic Noise Reduction by Frequency Domain Filtering

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Noise Models Restoration in the Presence of Noise - Only Spatial Filtering

- When an image is degraded only by additive noise,

$$g(x,y) = f(x,y) + \eta(x,y) \quad \text{in spatial domain}$$

$$G(u,v) = F(u,v) + N(u,v) \quad \text{in frequency domain}$$

- Spatial filtering is the method of choice for estimating f(x,y) [i.e., denoising image g(x,y)] in situations when only additive random noise is present.

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Mean Filters

Arithmetic Mean Filter

- The arithmetic mean filter is the same as the box filter
- Let S_{xy} represent the set of coordinates in a rectangular sub-image window (neighborhood) of size m×n ,centered on point (x,y).
- The arithmetic mean filter computes the average value of the corrupted image, g(x,y), in the area defined by S_{xy}.
- The value of the restored image $\hat{f}$ at point (x,y) is the arithmetic mean computed using the pixels in the region defined by S_{xy}.

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Mean Filters

Arithmetic Mean Filter

$$\hat{f}(x,y) = \frac{1}{mn} \sum_{(r,c)\in S_{xy}} g(r,c)$$

r and c are the row and column coordinates of the pixels contained in the neighborhood S_{xy}.

- This operation can be implemented using a spatial kernel of size m×n in which all coefficients have value 1/mn.
- A mean filter smooths local variations in an image, and noise is reduced as a result of blurring.

$$\frac{1}{9} \times \begin{array}{|c|c|c|}\hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline\end{array}$$

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Mean Filters

Geometric Mean Filter

- An image restored using a geometric mean filter is given by the expression

$$\hat{f}(x,y) = \left[\prod_{(r,c)\in S_{xy}} g(r,c) \right]^{\frac{1}{mn}}$$

where Π indicates multiplication.

- Each restored pixel is given by the product of all the pixels in the sub-image area, raised to the power 1/mn.
- A geometric mean filter achieves smoothing comparable to an arithmetic mean filter, but it tends to lose less image detail in the process.

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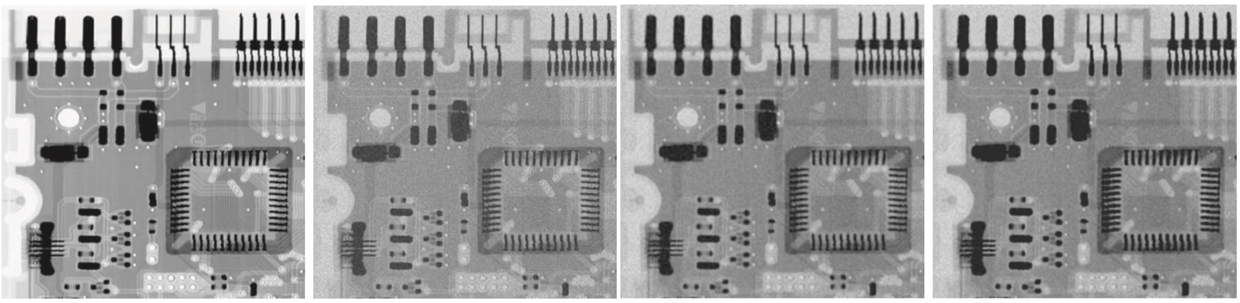
Mean Filters



(a) X-ray image of circuit board.
(b) Image corrupted by additive Gaussian noise.
(c) Result of filtering with an arithmetic mean filter of size 3×3 .
(d) Result of filtering with a geometric mean filter of the same size.

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Mean Filters

- Harmonic Mean
- The harmonic mean filtering operation is given by the expression

$$\hat{f}(x,y) = \frac{mn}{\sum_{(r,c)\in S_{xy}} \frac{1}{g(r,c)}}$$

- The harmonic mean filter works well for salt noise, but fails for pepper noise. It does well also with other types of noise like Gaussian noise.

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Mean Filters

Contra-harmonic Mean Filters

- The contra-harmonic mean filter yields a restored image based on the expression

$$\hat{f}(x,y) = \frac{\sum_{(r,c)\in S_{xy}} g(r,c)^{Q+1}}{\sum_{(r,c)\in S_{xy}} g(r,c)^{Q}}$$

where Q is called the order of the filter.

- The contra-harmonic filter is well suited for impulse noise, but it has the disadvantage that it must be known whether the noise is dark or light in order to select the proper sign for Q.
- The results of choosing the wrong sign for Q can be disastrous.

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Mean Filters

Contra-harmonic Mean Filters

- This filter is well suited for reducing or virtually eliminating the effects of salt-and-pepper noise.
  - For positive values of Q, the filter eliminates pepper noise.
  - For negative values of Q, it eliminates salt noise.
  - It cannot do both simultaneously.
- Note: The contra-harmonic filter reduces to the arithmetic mean filter if Q= 0, and to the harmonic mean filter if Q= –1.

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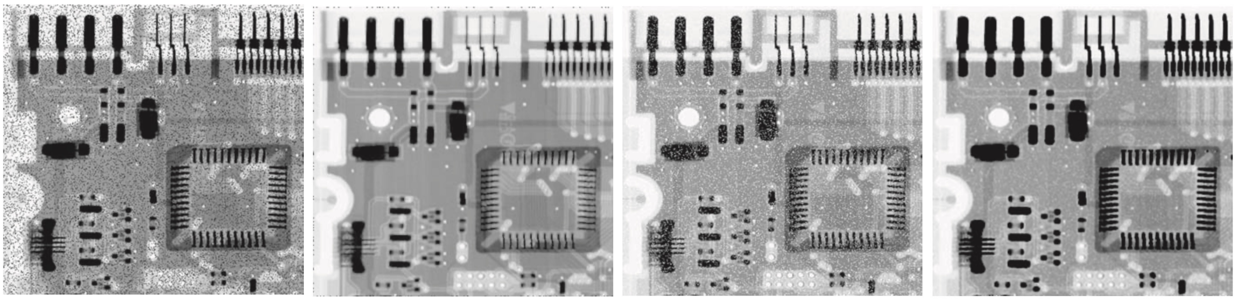
Mean Filters

Contra-harmonic Mean Filters



(a) Image corrupted by pepper noise with a probability of 0.1.
(b) Result of filtering (a) with a 3×3 contra-harmonic filter Q= 1.5
(c) Image corrupted by salt noise with the same probability.
(d) Result of filtering (b) with Q= −1.5

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Order Statistic Filters

- These are spatial filters whose response is based on ordering (ranking) the values of the pixels contained in the neighborhood encompassed by the filter.

Median Filter

- This filter replaces the value of a pixel by the median of the intensity levels in a predefined neighborhood of that pixel.

$$\hat{f}(x,y) = \underset{(r,c)\in S_{xy}}{\text{median}}\{g(r,c)\}$$

where, as before, Sxy is a subimage (neighborhood) centered on point (x,y).

- The value of the pixel at (x,y) is included in the computation of the median.

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Order Statistic Filters – Median Filter

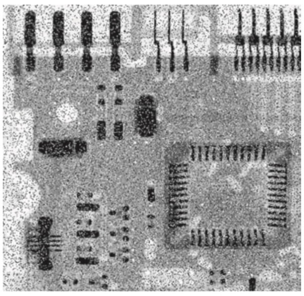
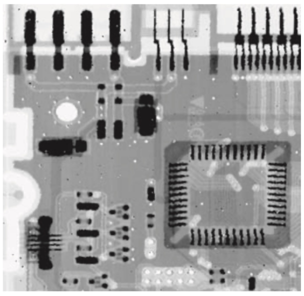
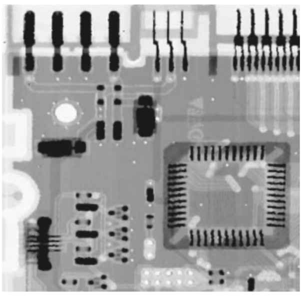


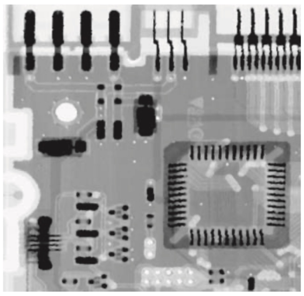
Image corrupted by salt-and-pepper noise with probabilities $P_s=P_p = 0.1$



The result of median filtering with a filter of size 3*3



Second pass



Third pass

These results are good examples of the power of median filtering in handling impulse-like additive noise. Keep in mind that repeated passes of a median filter will blur the image, so it is desirable to keep the number of passes as low as possible.

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Order Statistic Filters

Max Filters

- The median represents the 50th percentile of a ranked set of numbers, and using the 100th percentile results we get the max filter, given by

$$\hat{f}(x,y) = \max_{(r,c)\in S_{xy}} \{g(r,c)\}$$

- This filter is useful for finding the brightest points in an image or for eroding dark regions adjacent to bright areas.
- Also, because pepper noise has very low values, it is reduced by this filter as a result of the max selection process in the sub-image area S_{xy}.

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Order Statistic Filters – Max Filter

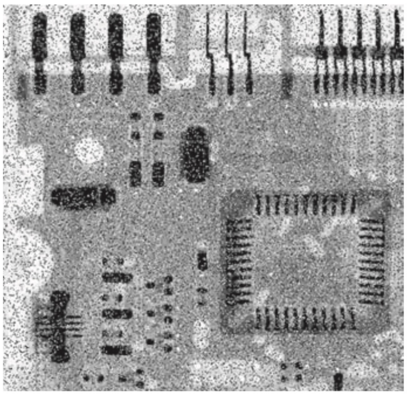
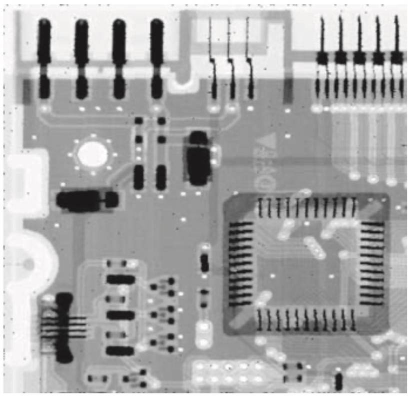


Image corrupted by salt-and-pepper noise with probabilities $P_s=P_p = 0.1$



Result of filtering with a max filter of size 3*3

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Order Statistic Filters

Min Filters

- The 0th percentile filter is the min filter:

$$\hat{f}(x,y) = \max_{(r,c)\in S_{xy}} \{g(r,c)\}$$

- This filter is useful for finding the darkest points in an image or for eroding light regions adjacent to dark areas.
- It reduces salt noise as a result of the min operation.

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Order Statistic Filters – Min Filter

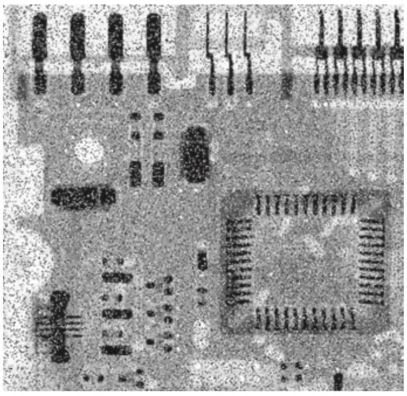
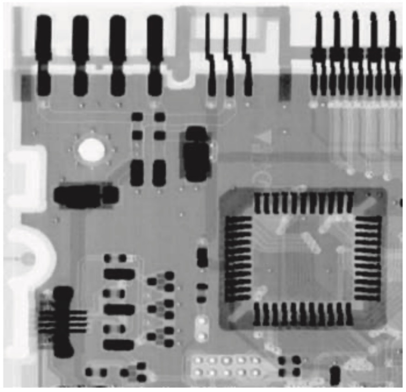


Image corrupted by salt-and-pepper noise with probabilities $P_s=P_p$ = 0.1



Result of filtering with a min filter of size 3*3

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Order Statistic Filters

Mid-Point Filter

- The midpoint filter computes the midpoint between the maximum and minimum values in the area encompassed by the filter:

$$\hat{f}(x,y) = \frac{1}{2}\left[\max_{(r,c)\in S_{xy}} \{g(r,c)\} + \min_{(r,c)\in S_{xy}} \{g(r,c)\}\right]$$

Note: This filter combines order statistics and averaging. It works best for randomly distributed noise, like Gaussian or uniform noise.

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Order Statistic Filters

Alpha-Trimmed Mean Filter

- Suppose that we delete the d/2 lowest and the d/2 highest intensity values of g (r,c) in the neighborhood S_{xy} and $g_R(r,c)$ represent the remaining mn−d pixels in S_{xy}.
- A filter formed by averaging these remaining pixels is called an alpha-trimmed mean filter.
- The form of this filter is $\hat{f}(x,y) = \frac{1}{mn-d} \sum_{(r,c)\in S_{xy}} g_R(r,c)$

where the value of d can range from 0 to mn−1.

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Order Statistic Filters

Alpha-Trimmed Mean Filter

- When d= 0 the alpha-trimmed filter reduces to the arithmetic mean filter.
- If we choose d = mn – 1, the filter becomes a median filter.
- For other values of d, the alpha-trimmed filter is useful in situations involving multiple types of noise, such as a combination of salt-and-pepper and Gaussian noise.

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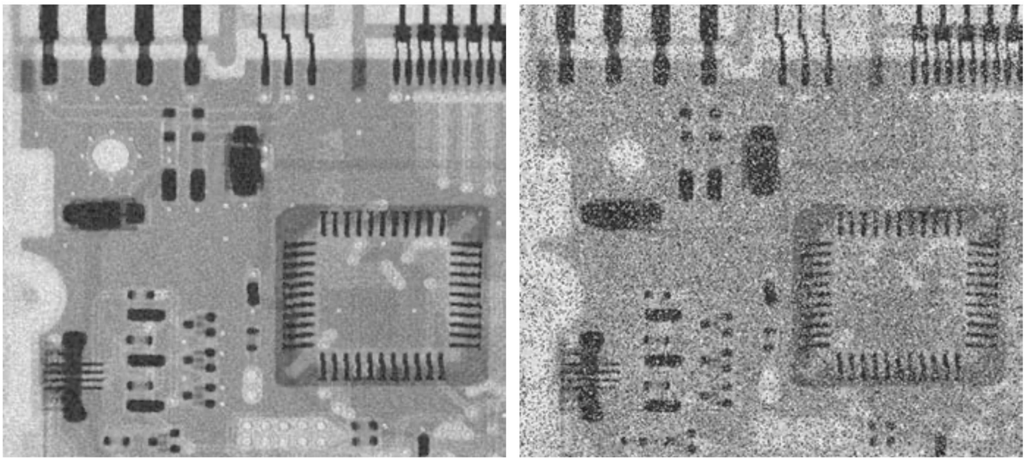
Impulse Noise Reduction using Spatial Filters – Examples



a) Image corrupted by additive uniform noise.
b) Image additionally corrupted by additive salt-and-pepper noise.

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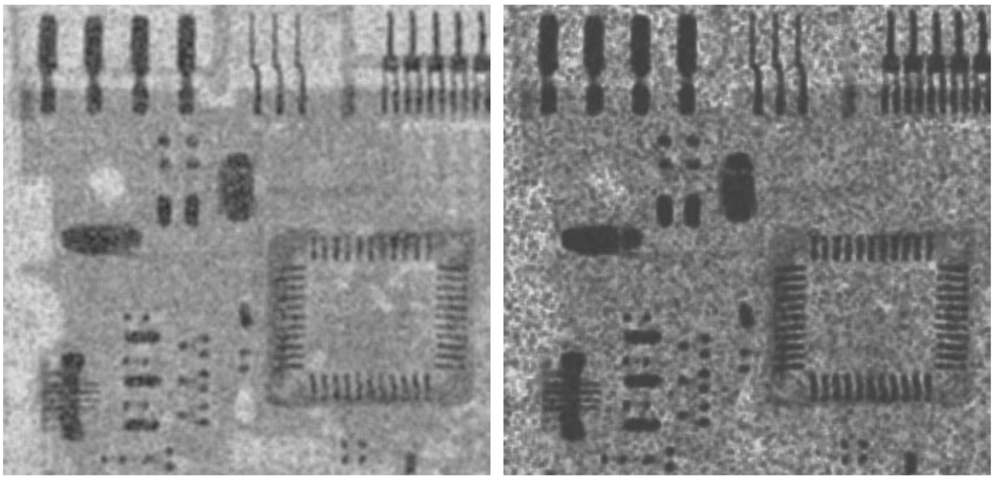
Impulse Noise Reduction using Spatial Filters – Examples



c) Applying arithmetic mean filter on image (b).
d) Applying geometric mean filter on image (b).

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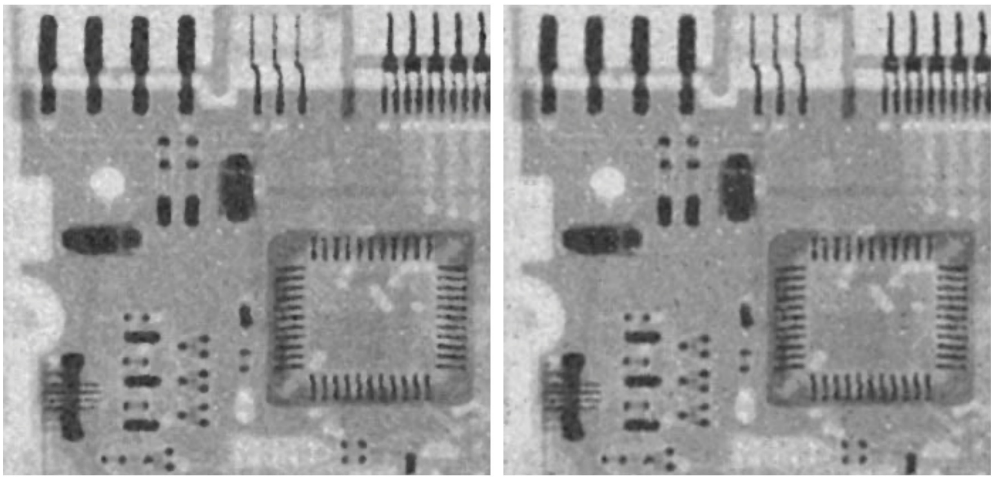
Impulse Noise Reduction using Spatial Filters – Examples



e) Applying median filter on image (b).
f) Applying alpha trimmed mean filter with d=6 on image (b).

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Adaptive Filters

- Adaptive filters are those whose behavior changes based on statistical characteristics of the image inside the filter region defined by the m×n rectangular neighborhood S_{xy}.

Adaptive - Local Noise Reduction Filters:

- Mean and variance are quantities closely related to the appearance of an image.
- The mean gives a measure of average intensity in the region over which the mean is computed, and the variance gives a measure of image contrast in that region.

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Adaptive – Local Noise Reduction Filter

- behavior of the filter to be as follows:
  1. If σ_n^2 is zero, the filter should return simply the value of g at (x,y). This is the trivial, zero-noise case in which g is equal to f at (x,y).
  2. If the local variance $\sigma_{S_{xy}}^2$ is high relative to σ_n^2, the filter should return a value close to g at (x,y). A high local variance typically is associated with edges, and these should be preserved.
  3. If the two variances are equal, we want the filter to return the arithmetic mean value of the pixels in S_{xy}. This condition occurs when the local area has the same properties as the overall image, and local noise is to be reduced by averaging.

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Adaptive – Local Noise Reduction Filter

- These filters operate on a neighborhood, Sxy , centered on coordinates (x,y).
- The response of the filter at (x,y) is to be based on:
  - g(x,y), the value of the noisy image at (x,y);
  - σ_n^2, the variance of the noise;
  - $\bar{z}_{S_{xy}}$, the local average intensity of the pixels in S_{xy} ; and
  - $\sigma_{S_{xy}}^2$, the local variance of the intensities of pixels in S_{xy}.
- An adaptive expression for obtaining f(x,y) based on these assumptions may be written as

$$\hat{f}(x,y) = g(x,y) - \frac{\sigma_\eta^2}{\sigma_{S_{xy}}^2}\left[g(x,y) - \bar{z}_{S_{xy}}\right]$$

σ_n^2 must be know priorly.

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Adaptive Median Filter

- Adaptive median filtering can handle noise with larger range of probabilities than normal median filter [range greater than 0.2].
- Also it seeks to preserve detail while simultaneously smoothing non-impulse noise, something that the "traditional" median filter does not do.
- The adaptive median filter also works in a rectangular neighborhood, S_{xy} unlike those filters
- However, the adaptive median filter changes (increases) the size of S_{xy} during filtering, depending on conditions as below:

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Adaptive Median Filter

- The adaptive median-filtering algorithm uses two processing levels, denoted level A and level B, at each point (x,y) :

Level A : If $z_{\text{min}} < z_{\text{med}} < z_{\text{max}}$, go to Level B
Else, increase the size of S_{xy}
If $S_{xy} \leq S_{\text{max}}$, repeat level A
Else, output z_{med}.

Level B : If $z_{\text{min}} < z_{xy} < z_{\text{max}}$, output z_{xy}
Else output z_{med}.

where
z_{min} = minimum intensity value in S_{xy}
z_{max} = maximum intensity value in S_{xy}
z_{med} = median of intensity values in S_{xy}
z_{xy} = intensity at coordinates (x, y)
S_{max} = maximum allowed size of S_{xy}

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Adaptive Median Filter – Simplified Process

1. Start with a small window (e.g., 3×3 around a pixel).
2. Find the median, the minimum, and the maximum pixel values in that window.
3. Check two things:
    1. Is the median value different from the minimum and maximum? → If yes, it's a good candidate.
    2. Is the center pixel different from the min and max? → If yes, keep it; otherwise, replace it with the median.
4. Still noisy?
    1. Increase the window size (e.g., from 3×3 to 5×5, up to a limit).
    2. Try again until a clean value is found or the maximum size is reached.

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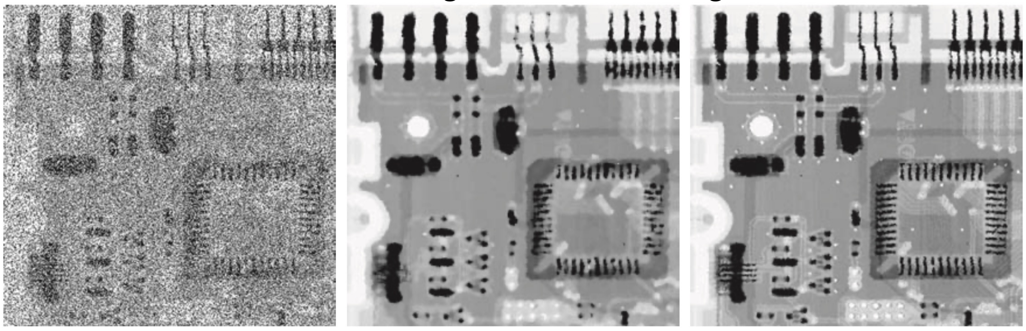
Adaptive Median Filter

- This produces a slightly less blurred result, but can fail to detect salt or pepper noise if that noise has the same brightness as the background."



(a)Image corrupted by salt-and-pepper noise with probabilities P_s=P_p=0.25
(b) Result of filtering with a 7×7 median filter.
(c) Result of adaptive median filtering with S_{max}= 7

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Periodic Noise Reduction using Frequency Domain Filtering

- periodic noise appears as concentrated bursts of energy in the Fourier transform, at locations corresponding to the frequencies of the periodic interference.
- The approach is to use a selective filter to isolate the noise
- The three types of selective filters
  - Band-reject,
  - Bandpass, and
  - Notch

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Selective Filtering

- Selective filtering is a technique used to modify or remove specific frequency components from an image.
- The main types of selective filters include:
  - **Bandpass Filtering** - Allows only a specific range of frequencies (a band) to pass through. Blocks frequencies below and above that band.
  - **Band Reject Filtering** - Removes a specific band of frequencies, while allowing low and high frequencies outside that band to pass.
  - **Notch Filtering** - A special case of band-reject filtering. Removes very narrow and localized frequency components (notches), rather than a wide band.

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Notch Filtering

- A notch filter rejects (or passes) frequencies in a predefined neighborhood of the frequency rectangle.
- Notch reject filter transfer functions are constructed as products of high-pass filter transfer functions whose centers have been translated to the centers of the notches. The general form is:

$$H_{\mathrm{NR}}(u,v) = \prod_{k=1}^{Q} H_k(u,v) H_{-k}(u,v)$$

- where $H_k(u,v)$ and $H_{-k}(u,v)$ are highpass filter transfer functions whose centers are at (u_k,v_k) and $(-u_k,-v_k)$ respectively.
- These centers are specified with respect to the center of the frequency rectangle, (M/2,N/2) where, as usual, M and N are the number of rows and columns in the input image.

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Notch Filtering

- Thus, the distance computations for each filter transfer function are given by:

$$D_k(u,v) = \left[(u - M/2 - u_k)^2 + (v - N/2 - v_k)^2\right]^{1/2}$$

$$D_{-k}(u,v) = \left[(u - M/2 + u_k)^2 + (v - N/2 + v_k)^2\right]^{1/2}$$

- Butterworth notch reject filter transfer function of order n, containing three notch pairs:

$$H_{\mathrm{NR}}(u,v) = \prod_{k=1}^{3} \left[\frac{1}{1 + \left[D_{0k}/D_k(u,v)\right]^n}\right]\left[\frac{1}{1 + \left[D_{0k}/D_{-k}(u,v)\right]^n}\right]$$

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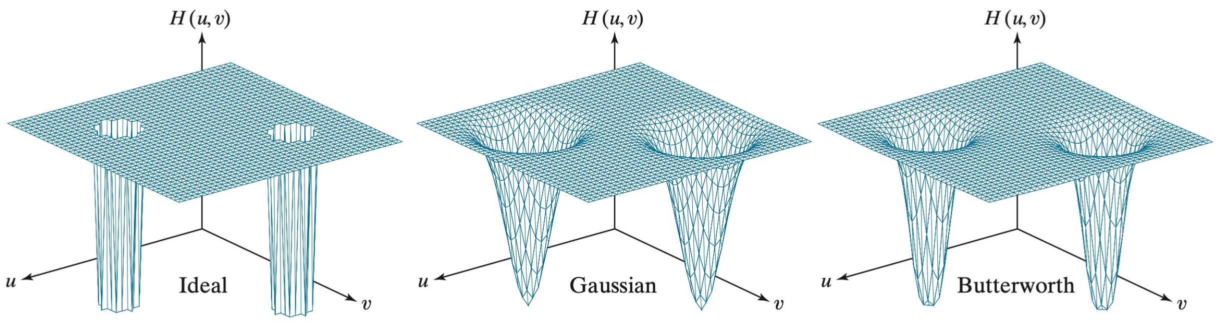
Notch Filtering



Perspective plots of (a) ideal, (b) Gaussian, and (c) Butterworth notch reject filter transfer functions.

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Notch Filtering

- A notch pass filter transfer function is obtained from a notch reject function using the expression $H_{\mathrm{NP}}(u,v) = 1 - H_{\mathrm{NR}}(u,v)$

a) Sample Notch pass filter function
b) Spatial pattern obtained by computing the IDFT of (a).

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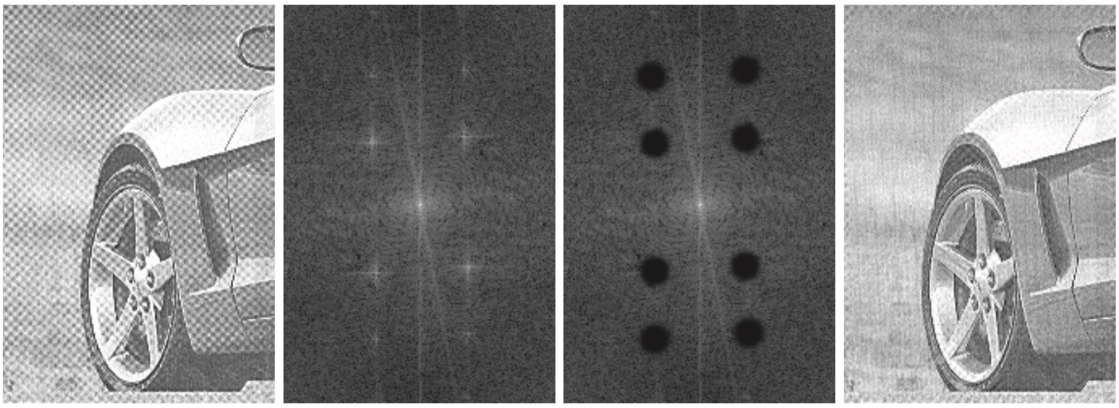
Notch Filtering



(a) Sampled newspaper image showing a moiré pattern.
(b) Spectrum.
(c) Fourier transform multiplied by a Butterworth notch reject filter transfer function.
(d) Filtered image.

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Optimum Notch Filtering

- When noise is complex, semi-periodic, or not limited to sharp frequency spikes standard notch filtering (which often uses fixed heuristics to block interference) cannot perform well but optimum notch filtering adapts to the local structure of the interference for more accurate and less destructive restoration.

1. **Extract interference frequency components**

   Construct a notch pass filter $H_{NP}(u,v)$ to isolate spikes in the Fourier domain that correspond to periodic noise:

   $$\mathbf{N(u,v)=H_{NP}(u,v)\cdot G(u,v)}$$

   G(u,v) is the DFT of the noisy image.
   N(u,v) is the estimated interference component in the frequency domain.

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Optimum Notch Filtering

2. Inverse Transform to Spatial Domain

$$h(x,y) = \mathfrak{F}\{N(u,v)\}$$

This gives an approximation of the spatial interference pattern $\eta(x,y)$.

3. Weighted Subtraction

$$\hat{f}(x,y)=g(x,y)-w(x,y)\cdot \eta(x,y)$$

Where g(x,y) observed noisy image.

$\hat{f}(x,y)$ restored image.

w(x,y): local weight to be computed such that the local variance of $\hat{f}(x,y)$ is minimized.

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Optimum Notch Filtering

4. Minimizing Local Variance

- Use a neighborhood S_{xy} around pixel (x,y) and define local variance as: $\sigma^2(x,y) = \frac{1}{mn}\sum_{(r,c)\in S_{xy}} [\hat{f}(r,c) - \bar{\hat{f}}]^2$

  $\bar{\hat{f}}$ is the average value of $\hat{f}$ in neighborhood S_{xy}.
- Plug in the weighted subtraction formula:

  $$\hat{f}(r,c)=g(r,c)-w(x,y)\cdot \eta(r,c)$$
- Then differentiate $\sigma^2(x,y)$ with respect to w(x,y) and set derivative to zero. The optimal weight is: $w(x,y) = \frac{\bar{g}.\bar{\eta}}{\bar{\eta}^2}$

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Optimum Notch Filtering

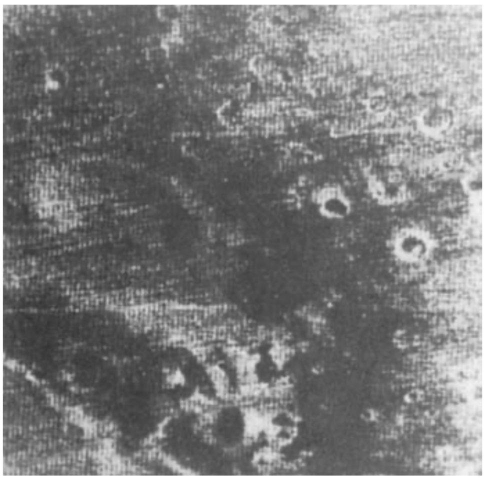
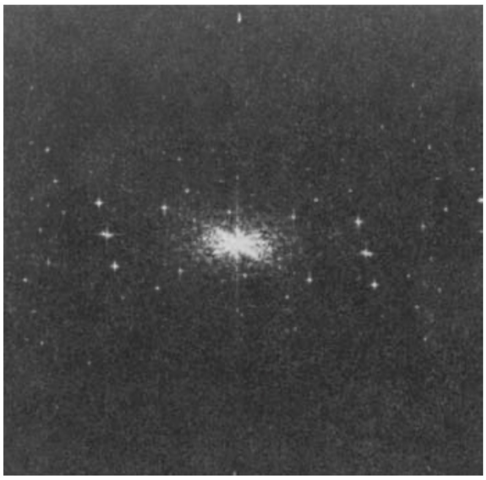


Image of the Martian terrain taken by Mariner 6.



Fourier spectrum showing periodic interference.

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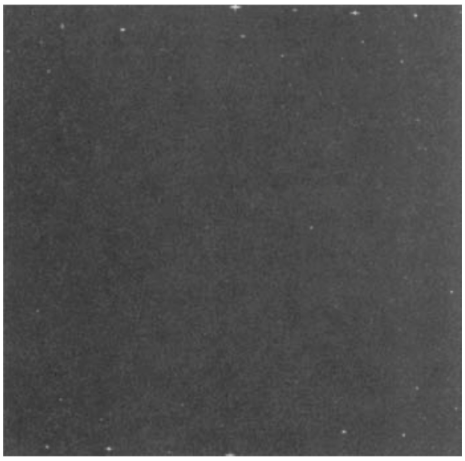
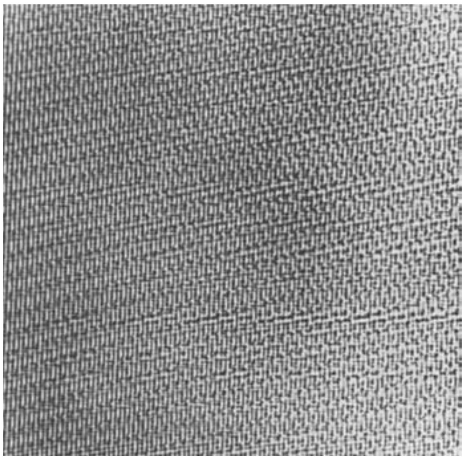
Optimum Notch Filtering



Fourier spectrum of N(u,v)



Corresponding spatial noise interference pattern

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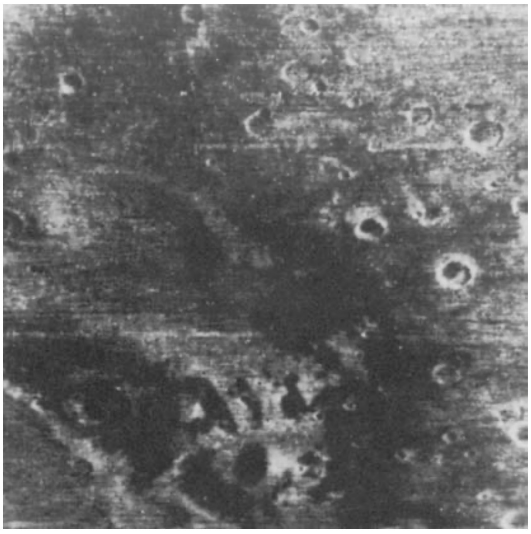
Optimum Notch Filtering



Restored Image

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Discussions and Classwork

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